

$f(x)$

Module 4

Chapter 6

Gaussian and the Central Limit Theorem

What is gaussian Distribution?

Gaussian distribution, also known as the **Normal distribution**, is a **probability density function** which comes up quite a bit in natural settings. The curve traced by the upper values of the normal Distribution is in the shape of a bell, hence it is also called the “**Bell Curve**” (Fig 1). Not only are biological measurements like height and weight often “normally distributed,” but sensor error and measurement error also often follows this distribution. That’s because, as we’ll see when we discuss the Central Limit Theorem, the Gaussian is what you get when you **sum many independent and identically distributed random variables**. Other processes can result in other distributions, but this idea of many small factors contributing to a number is just very widely applicable; and even when they aren’t quite identically distributed, you can still end up with the Gaussian being a good model for the distribution.

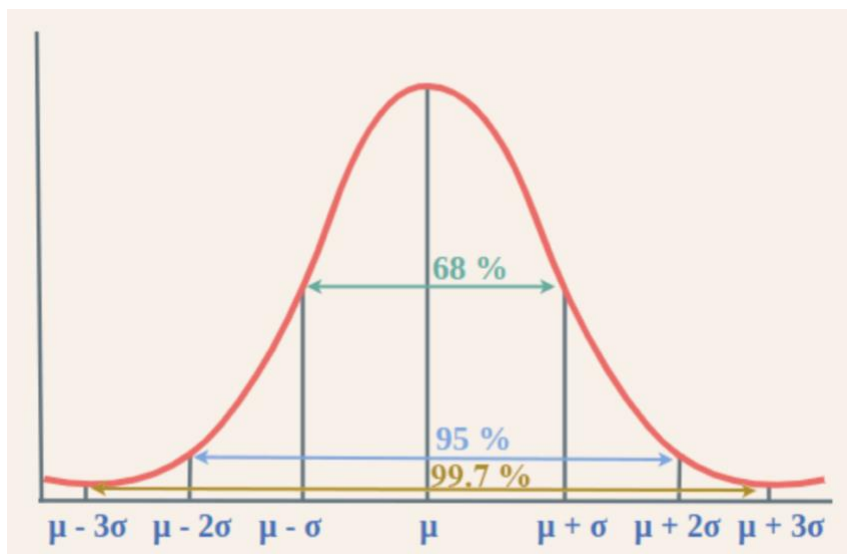


Fig 1 (Picture Source: <https://www.geeksforgeeks.org/normal-distribution/>)

Normal distributions can differ in their means and in their standard deviations. The figure below (Fig 2) shows three normal distributions. The green (left-most) distribution has a mean of -3 and a standard deviation of 0.5, the distribution in red (the middle distribution) has a mean of 0 and a standard deviation of 1, and the distribution in black (right-most) has a mean of 2 and a standard deviation of 3.

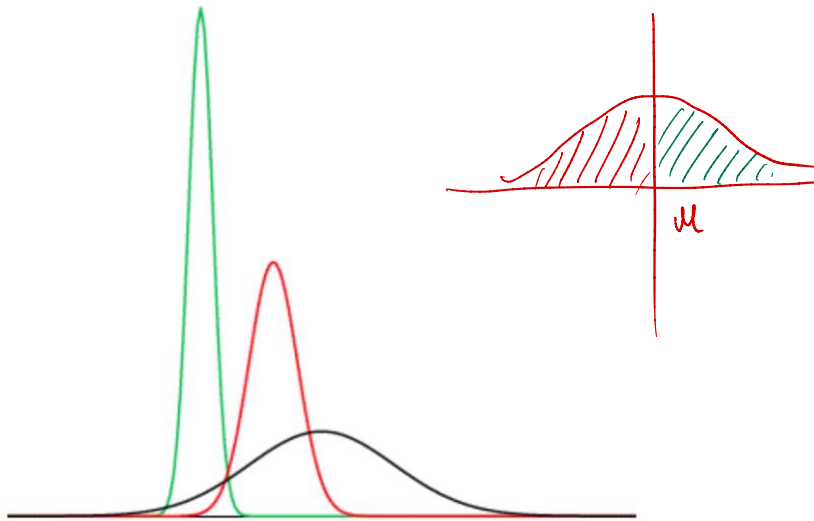


Fig 2 (Picture Source: <https://www.inprolink.com/blog/what-is-normal-distribution/>)

Following are the most important properties of a Gaussian Distribution:

- Normal distributions are ***symmetric*** around their ***mean***.
- The mean, median, and mode of a normal distribution are equal.
- The ***area*** under the normal curve is equal to ***1.0***.
- Normal distributions are denser in the center and less dense in the tails.
- Normal distributions are defined by two parameters, the ***mean (μ)*** and the ***standard deviation (σ)***.
- ***68%*** of the area of a normal distribution is within ***one standard deviation of the mean***.

- Approximately **95%** of the area of a normal distribution is within two *standard deviations of the mean.*

1.96 s.d away

**The density of the normal distribution (the height for a given value on the x axis) is given by :

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Law of Large Numbers

Let's start with two examples:

Example 1: The coin Toss:

Imagine a situation where you toss a fair coin twice. What would you expect to happen? You'd probably expect to get 1 head & 1 tail. However, things may go a little differently and you may get 2 heads or 2 tails. The latter 2 outcomes are less likely, but you probably won't be too surprised if that happens. And why would you? It's just two coins :)

Now, if you toss that same coin 20 times, you'd expect around 10 tails and 10 heads. This is all under the assumption that the coin is fair. Again, the actual results may be different, so you may end up with, for instance, 8 tails and 12 heads. Alternatively, you may end up with 1 tail and 19 heads, although this is a highly unlikely scenario. And if this ever happens, you should suspect the fairness of the coin.

Let's now imagine tossing the same coin 1000 times. How many tails would you expect to get? Likely around 500. Maybe 450, maybe 510 —but probably very close to 500. You may end up with 10 heads and 990 tails, but this is quite improbable. You'd be quite surprised if this happens. So the more we toss the same fair coin, the more likely it is that the **actual** ratio of *heads* : *tails* will be close to the **expected** ratio of 50–50. This is perfectly intuitive. And again, this is all under the assumption that the coin is fair. Otherwise, if you had a biased coin with 90%

chances of getting a tail, you'd expect to get a tail most of the times when tossing the coin.

So what does all of this mean to us?

The reality will meet the expectations more closely when we keep on increasing the number of times we run an experiment.

Let's consider another example.

Example 2: The Dice Roll

Now let's grab a fair dice instead of a coin, and roll it. If we roll a dice many, many times, the expected average of these rolls is going to be 3.5, i.e. $(1 + 2 + \dots + 6) / 6$. The more we roll the dice, the more stable this expectation is, and the average won't deviate much from 3.5. At the very beginning, the average outcome will be quite unstable, but later on it will stabilize around 3.5.

Now we understand what happens when we repeat an experiment more and more times. The reality begins to meet the expectations. In other words, the **actual average** closely resembles the **expected average**.

To formalize this concept in a more mathematical way, let $X_1, X_2, \dots$ be a finite or infinite sequence of random variables, each with the same expectation μ :

$E(X_1) = E(X_2) = \dots = \mu$. In the Coin Toss example, *this condition ensures that the coin tosses are fair, with the same expected values.*

According to the **LLN**,

^{observed}
The sample mean $\bar{X} = \frac{(X_1 + X_2 + X_3 + \dots)}{n}$ *converges to μ as n converges to infinity*

(i.e. as the number of trials keeps increasing).

Now you have an intuition of what the Law of Large Numbers means to us. It has numerous real-life applications — e.g. a casino may lose money from a single player that got lucky, but in the long run, it will gain money from many other players who were unfortunate to lose. Ultimately, the statisticians working at this

casino are actually fully aware of the LLN — they intentionally optimize the game rules so that over the long run, the wins and losses will cluster around a predictable ratio. This way the casino's earnings are maximized.

Now, since all the X_i 's are random variables, the Sample Mean, $\bar{X}$ is also a random variable. Thus, by linearity of expectation, we have, **Mean** of $\bar{X}$:

$$E[\bar{X}] = E\left\{\frac{(X_1 + X_2 + X_3 + \dots + X_n)}{n}\right\}$$

$E\left\{\frac{1}{n}\{X_1 + X_2 + \dots + X_n\}\right\}$
 $\frac{1}{n} E\{X_1 + X_2 + \dots + X_n\}$

$$E[\bar{X}] = \frac{E[X_1] + E[X_2] + E[X_3] + \dots + E[X_n]}{n}$$

$$E[\bar{X}] = \frac{nE[X]}{n} = \frac{n\mu}{n} = \mu$$

$Var[aX] = a^2 Var[X]$

The **variance** of $\bar{X}$ is given by:

$$Var[\bar{X}] = Var\left\{\frac{(X_1 + X_2 + X_3 + \dots + X_n)}{n}\right\}$$

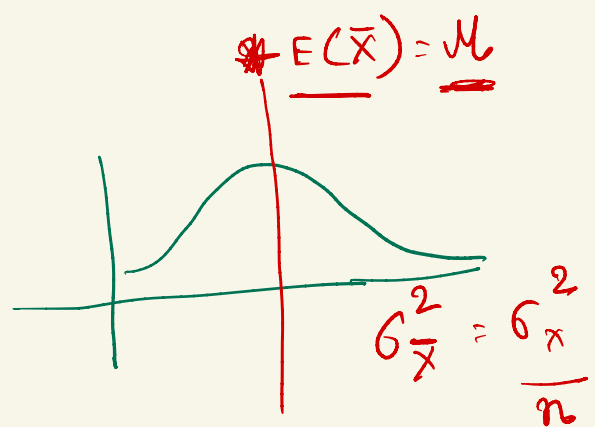
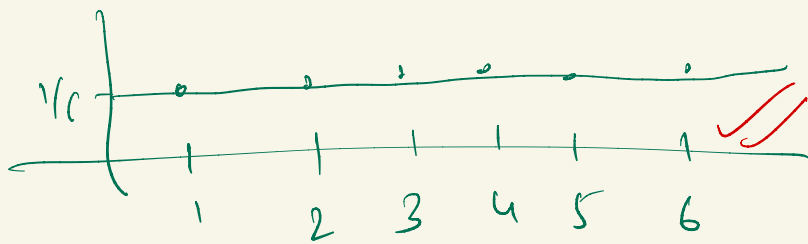
$$Var[\bar{X}] = \frac{Var[X_1] + Var[X_2] + Var[X_3] + \dots + Var[X_n]}{n^2} \rightarrow \frac{\sigma^2}{n}$$

$\frac{\sigma^2}{n^2}$

$$\underline{Var[\bar{X}]} = \frac{nVar[X]}{n^2} = \frac{Var[X]}{n} = \frac{\sigma^2}{n}$$

Central Limit Theorem (CLT)

The **central limit theorem (CLT)** is one of the most important results in probability theory. There are two alternative forms of the theorem, and both alternatives are concerned with drawing finite **samples size n** from a population with a known **mean, μ** , and a known **standard deviation, σ** . The first alternative says that if we collect samples of size n with a "**large enough n** ," calculate each **sample's mean**, and create a **histogram of those means**, then the resulting histogram will tend to have an **approximate normal bell shape**. The second alternative says that if we again collect samples of size n that are "**large enough**," calculate the **sum of each**



Sample 1 → $n = 35$
 $\bar{X}_1 = 4$

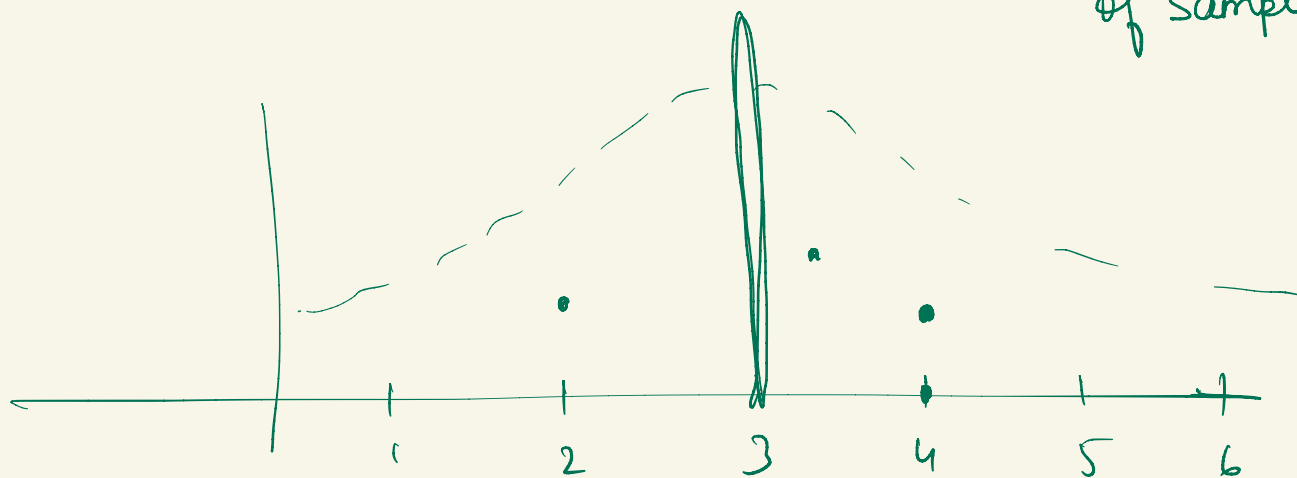
Sample 2 → $n = 35$
 $\bar{X}_2 = 2$

Sample 3 → $n = 35$
 $\bar{X}_3 = 3.6$

$\sigma_{\bar{X}} = \frac{\sigma_x}{\sqrt{n}}$

$n \geq 30$

Sampling distribution
of Sample Means



sample and create a *histogram*, then the resulting histogram will again tend to have a *normal bell-shape*.

*****In either case, it does not matter what the distribution of the original population is, or whether you even need to know it. The important fact is that the distribution of sample means and the sums tend to follow the normal distribution.***

The *size of the sample, n*, that is required in order to be "large enough" depends on the original population from which the samples are drawn (the sample size should be at *least 30* or the data should come from a normal distribution). If the original population is far from normal, then more observations are needed for the sample means or sums to be normal. **Sampling is done with replacement.**

CLT for Sample Means

Suppose X is a random variable with a distribution that may or may not be known. Using subscript that matches the random variable, let:

$\mu_x = \text{Mean of } X$

$\sigma_x = \text{Standard Deviation of } X$

If you draw random *samples of size n*, then as n increases, the random variable $\bar{X}$ which consists of sample means, tends to be normally distributed and the normal distribution has the *same mean* as the *original distribution* (μ_x) and a *variance* that equals the *original variance divided by n*, the sample size $\left(\frac{\sigma_x^2}{n}\right)$. The variable n is the *number of values that are averaged together, not the number of times the experiment is done*.

Formally speaking,

If you draw random samples of size n , the distribution of the random variable $\bar{X}$, which consists of sample means, is called the *sampling distribution of the mean*.

$$P(\bar{X}' < \bar{X})$$

The sampling distribution of the mean approaches a normal distribution as n , the sample size, increases.

The random variable $\bar{X}$ has a *z-score* associated with it which is given by:

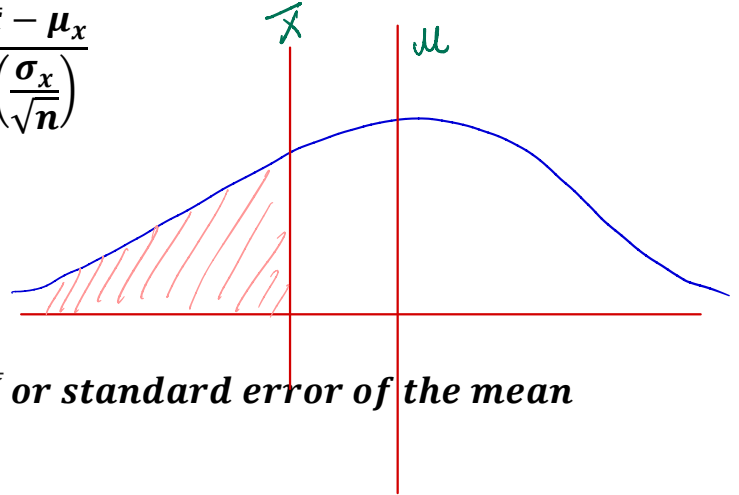
$$z = \frac{\bar{X} - \mu_x}{\sigma_{\bar{X}}}$$

$$z = \frac{\bar{x} - \mu_x}{\left(\frac{\sigma_x}{\sqrt{n}}\right)}$$

Here,

μ_x = mean of X and $\bar{X}$

$\frac{\sigma_x}{\sqrt{n}} = \sigma_{\bar{x}}$ = Standard deviation of $\bar{X}$ or standard error of the mean



The *z-score* tells you *how many standard deviations away from the mean* a given observation is while a *z-table* shows the percentage or *probability of a value* being *less than a given z-score* in a standard normal distribution.

↓
area under the curve
for a given z-score

Let's take an example

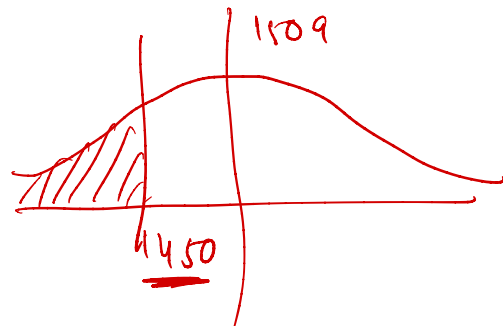
In 2019, the mean score for three sections of the SAT was $\mu = 1509$. The standard deviation was $\sigma = 339$. Let's answer the following questions:

- 100 students take the SAT. What is the expected value of their average score and what is the standard error of the sampling distribution?
- What is the probability that the average score of those 100 students will be less than 1450?
- What is the probability that the average score of the 100 students is between 1450 and 1600?

Let's answer the first part:

Using CLT, for sample size $n = 100$,

$$P(X < 1450)$$



$$E(\bar{X}) = \mu$$

$$\frac{\sigma_x}{\sqrt{n}}$$

$$E[\bar{X}] = \mu_x = 1509$$

Standard error is given by:

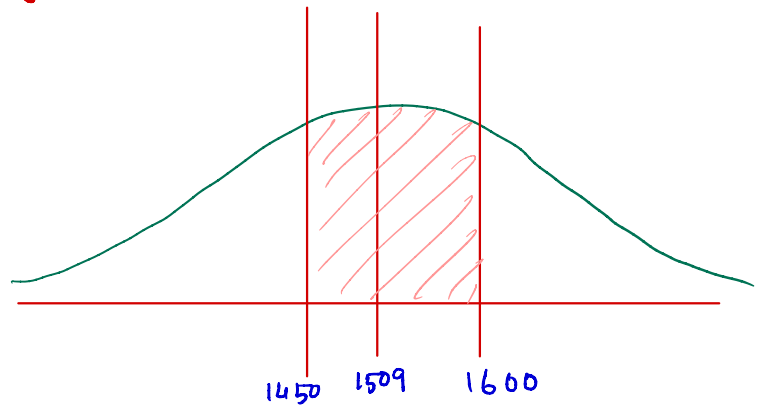
$$\frac{\sigma_x}{\sqrt{n}} = \frac{339}{\sqrt{100}} = 33.9$$

Let's answer the second part:

$$z_{1450} = \frac{1450 - 1509}{33.9} \approx -1.74, P(\bar{X} < 1450) = P(Z < -1.74) = 0.0409$$

Let's now answer the third part:

$$z_{1600} = \frac{1600 - 1509}{33.9} \approx 2.68, P(1450 < \bar{X} < 1600) = P(Z < 2.68) - P(Z < -1.74) = 0.9963 - 0.0409 = 0.9554$$



$$P(\bar{X} > 110)$$

Let's Practice

- You pay \$1 to play a game in which you roll one standard six-sided die. You lose your dollar if the die is 1, 2, 3 or 4. You get your dollar back if the die is a 5, and if the die is a 6 you get your dollar back plus \$2 more (total of \$3).
 - Calculate expected value and standard deviation for a single toss.
 - If you play the game 100 times, what are the expected value and standard error for the sampling distribution?
 - If you play the game 100 times, what is the probability that your average outcome will be positive?
 - If you play the game 100 times, your average winnings have a 90% probability of being below what value?
- The average human drinks 2L of water when active outdoors, with a standard deviation of 0.7L. You are planning a full day of nature trip for 50 people and will bring 110L of water. What is the probability that you will run out?
- A large elevator can transport a maximum of 9800lbs. Suppose a load of cargo containing 49 boxes must be transported via the elevator. Experience has shown the weight of boxes of this type of cargo follow a distribution with $\mu = 205\text{lbs}$ and $\sigma = 15\text{lbs}$. Based on this information, what is the probability that all 49 boxes can be safely loaded onto the elevator and transported?

- An airplane has a capacity of 200 seats and a total baggage limit of 6000kgs. Assume that the total weight X checked in by each passenger is a random variable with a mean of 28kgs and a standard deviation of 15kgs. If 200 passengers board the flight, what is the probability that the total weight of their baggage will not exceed the limit?
- A survey found that an American family generates on an average of 17.2lbs of glass garbage each year. Assume the standard deviation of the distribution to be 2.5lbs. Find the probability that the mean of a sample of 55 families will be between 17lbs and 18lbs.
- The average teacher's salary in NJ is \$52,174. Suppose the distribution is normal with a standard deviation of \$7500.
 - a) What is the probability that a randomly chosen teacher has a salary less than \$50,000 per year?
 - b) If we sample 100 teachers' salaries, what is the probability that the mean salary is less than \$50,000 per year?

1.

- a) Creating a probability distribution table for a Random Variable X that represents the game

X	-1	0	2 (\$3 - \$1)
$P(X=x)$	$\frac{4}{6} = \frac{2}{3}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E(X) = \mu_X = -1\left(\frac{2}{3}\right) + 0 \times \frac{1}{6} + 2 \times \frac{1}{6} = -\frac{1}{3}$$

$$\begin{aligned} \text{Var}(X) = \sigma_X^2 &= \left(-1 + \frac{1}{3}\right)^2 \times \frac{2}{3} + \left(0 + \frac{1}{3}\right)^2 \times \frac{1}{6} + \left(2 + \frac{1}{3}\right)^2 \times \frac{1}{6} \\ &= \frac{11}{9} \approx 1.222 \end{aligned}$$

$$\text{s.d.}(X) = \sigma_X = \sqrt{\frac{11}{9}} \approx 1.1055 \approx 1.106$$

- b) By CLT, $n = 100$

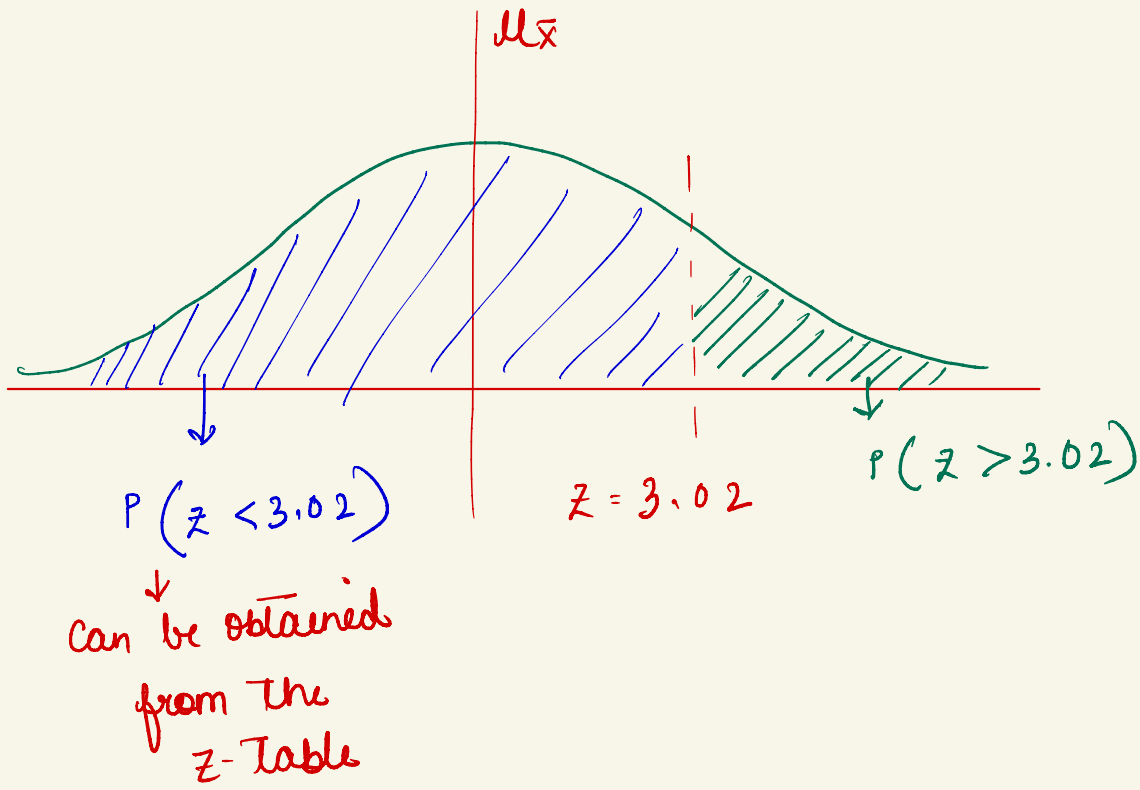
$$\therefore E(\bar{X}) = \mu_X = -\frac{1}{3}$$

$$\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}} \approx \frac{1.106}{\sqrt{100}} = 0.1106$$

- c) Probability that my average outcome is positive
 $= P(\bar{X} > 0)$

$$Z = \frac{\bar{X} - \mu_X}{\sigma_X/\sqrt{n}} = \frac{0 - (-1/3)}{0.1106} \approx 3.02$$

$$\therefore P(X > 0) = P(Z > 3.02) = 1 - P(Z < 3.02) = 1 - 0.9987 = 0.0013$$



d) we need to find the z-score associated with a value closest to 0.90 in the z-table.

The closest value is 0.8997. The associated z-score is 1.28

$$\therefore 1.28 = \frac{\bar{x} - \mu_x}{\sigma_x / \sqrt{n}} = \frac{\bar{x} + 1/3}{\sqrt{11/900}} = \frac{\bar{x} + 1/3}{\sqrt{11}/30}$$

$$\Rightarrow \bar{x} = \left\{ 1.28 \times \left(\frac{\sqrt{11}}{30} \right) \right\} - 1/3 \approx -0.19$$

2. Given,

$$\mu_x = 2 = \mu_{\bar{x}}, \quad n = 50$$

$$\sigma_x = 0.7 \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = 0.7/\sqrt{50} = 0.099$$

$$P(\text{we will run out}) = P(\text{use more than 110L})$$

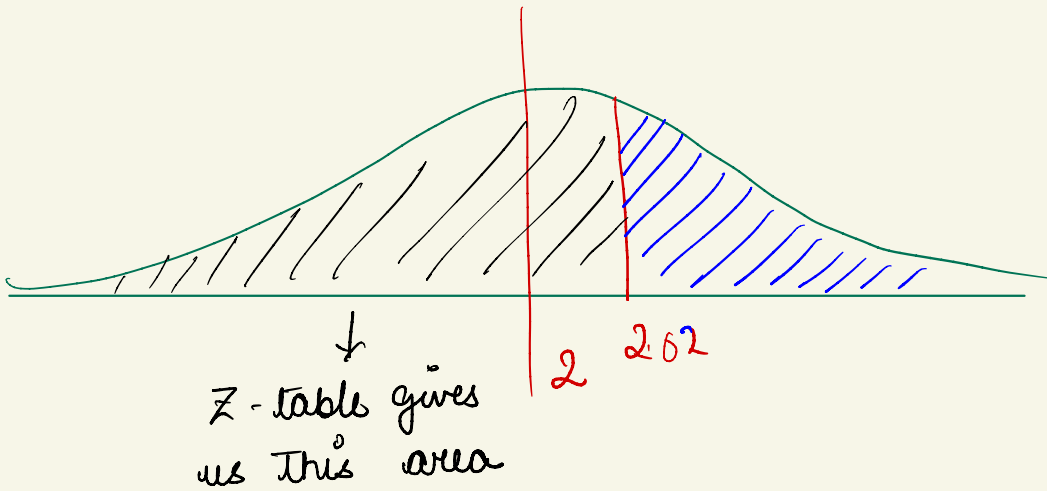
$$\bar{x} = \frac{110}{50} = 2.2 \text{ L}$$

$$\therefore P(\text{av. consumption of water per man} > 2.2 \text{ L})$$

$$\therefore P(x > 2.2 \text{ L})$$

$$Z = \frac{2.2 - 2}{0.099} = 2.02$$

$$P(Z \leq 2.02) = 0.9783$$



we are interested in $P(Z > 2.02)$

$$\Rightarrow 1 - P(Z < 2.02)$$

$$= 1 - 0.9783$$

$$= 0.0217$$

$\Rightarrow$ There is a 2.17% chance that we will run out of water.

3. $n = 49$

$$\mu_x = 205$$

$$\sigma_x = 15$$

The elevator can transport 9800 lbs. Therefore, these 49 boxes can be safely transported if they weigh less than $\frac{9800}{49}$ lbs ≈ 200 lbs

we are looking at $P(\bar{x} < 200)$

$$\Rightarrow Z_{200} = \frac{200 - 205}{15/\sqrt{49}} = \frac{-5 \times 7}{15}$$

$$= -2.33$$

$$\therefore P(Z < -2.33) = 0.0099$$

4. For the luggage of 200 passengers to not exceed 6000 kgs, the average weight checked by each passenger must not exceed $\frac{6000}{200} = 30$ kgs

$\therefore$ we are interested in finding

$$P(\bar{X} \leq 30)$$

given

$$n = 200$$

$$\mu_{\bar{X}} = \mu_X = 28, \quad \sigma = 15$$

$$\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}} = \frac{15}{\sqrt{200}}$$

$$Z_{30} = \frac{30 - 28}{\frac{15}{\sqrt{200}}} \approx 1.88561$$

$$\therefore P(\bar{X} \leq 30) = P(X < 1.88561) \approx 0.9703 \\ = 97.03\%$$

5.

Given,

$$\mu_{\bar{x}} = 17.2$$

$$n = 55$$

$$\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}} \quad \left(\sigma_{\bar{x}}^2 = \frac{\sigma_x^2}{n} \Rightarrow \sqrt{\sigma_{\bar{x}}^2} = \frac{\sigma_x}{\sqrt{n}} \right) = 2.5 / \sqrt{55} = 0.3371$$

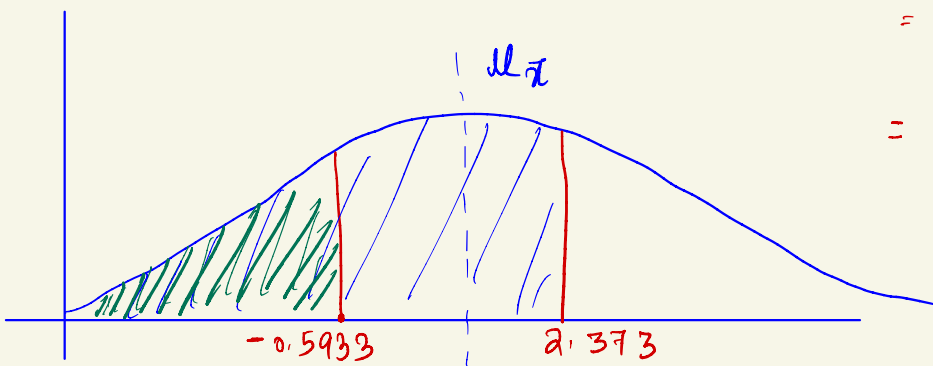
we are interested in $P(17 < \bar{x} < 18)$

$$Z_{17} = (17 - 17.2) / 0.3371 = -0.5933$$

$$Z_{18} = (18 - 17.2) / 0.3371 = 2.373$$

This is where we use the z-table:

$$\begin{aligned} P(-0.5933 < Z < 2.373) \\ &= 0.9911 - 0.2776 \\ &= 0.7135 \end{aligned}$$



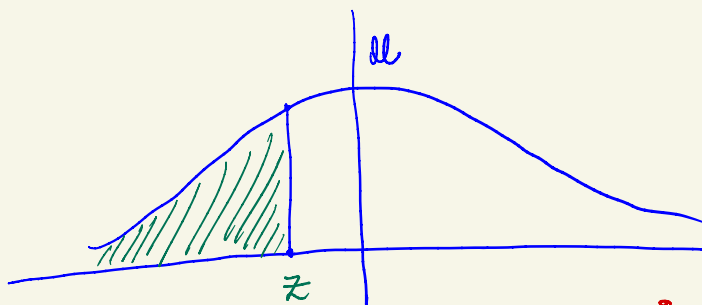
6) a. $\mu_x = 52174$
 $\sigma_x = 7500$

Since the distribution is normal, our samples follow a normal distribution.

$$\therefore Z_{50,000} = \frac{\bar{x} - \mu_x}{\sigma} = \frac{50000 - 52174}{7500}$$

$$= -0.2899$$

$$\therefore P(x < 50000) = P(Z < -0.2899) = 0.3860$$



b. $\mu_x = 52174$
 $\sigma_x = 7500$

Here we are looking at sampling distribution of sample means.

$$\therefore \sigma_{\bar{x}} = \frac{7500}{\sqrt{n}}$$

Here $n = 100$

$$\therefore \sigma_{\bar{x}} = \frac{7500}{\sqrt{100}} = 750$$

$$\therefore Z_{50,000} = \frac{50,000 - 52174}{750} = -2.8987$$

$$\therefore P(\bar{x} < 50,000) = P(Z < -2.8987) = 0.0019$$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990